

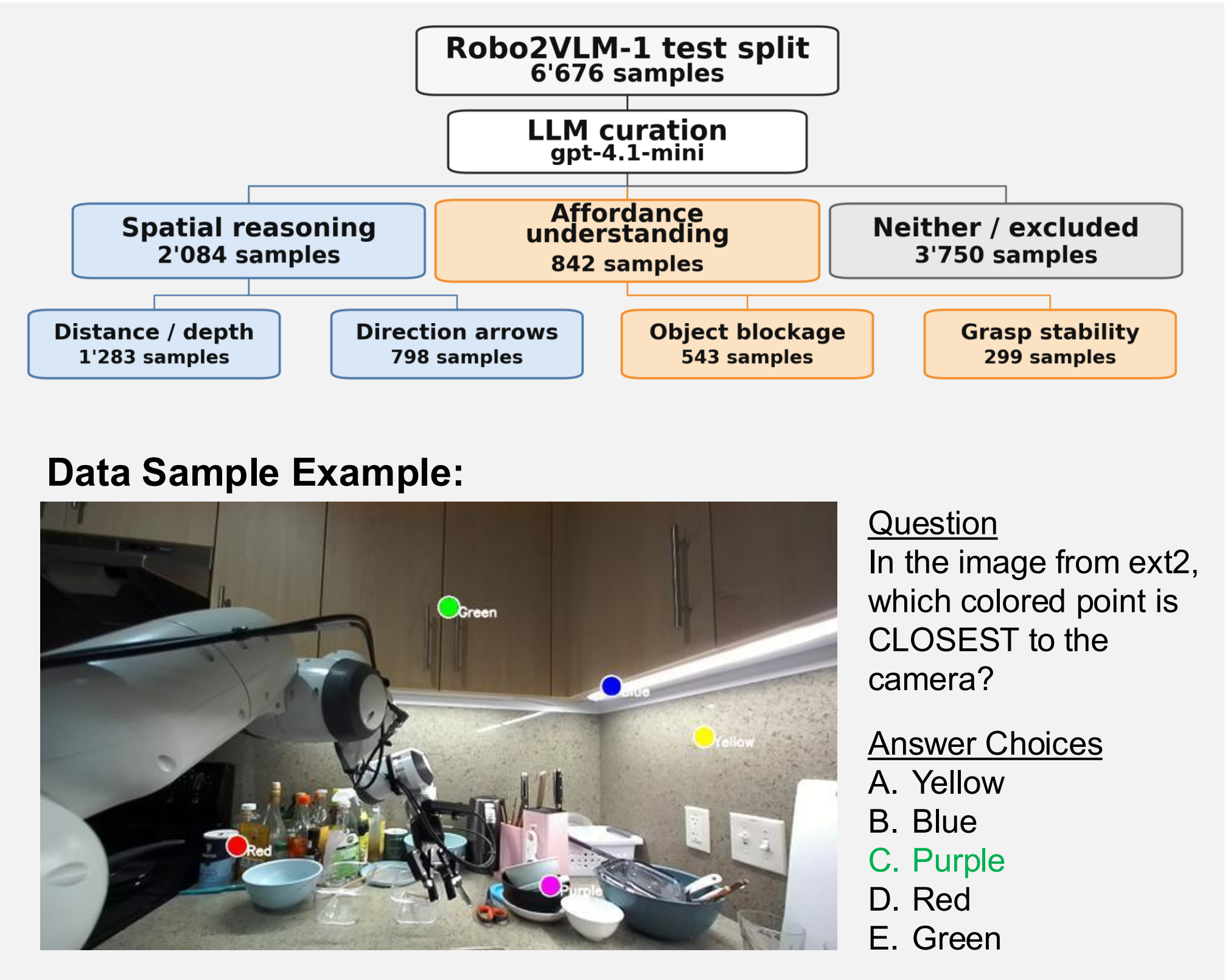
Benchmarking Spatial and State Reasoning Skills of VLMs for Robotics

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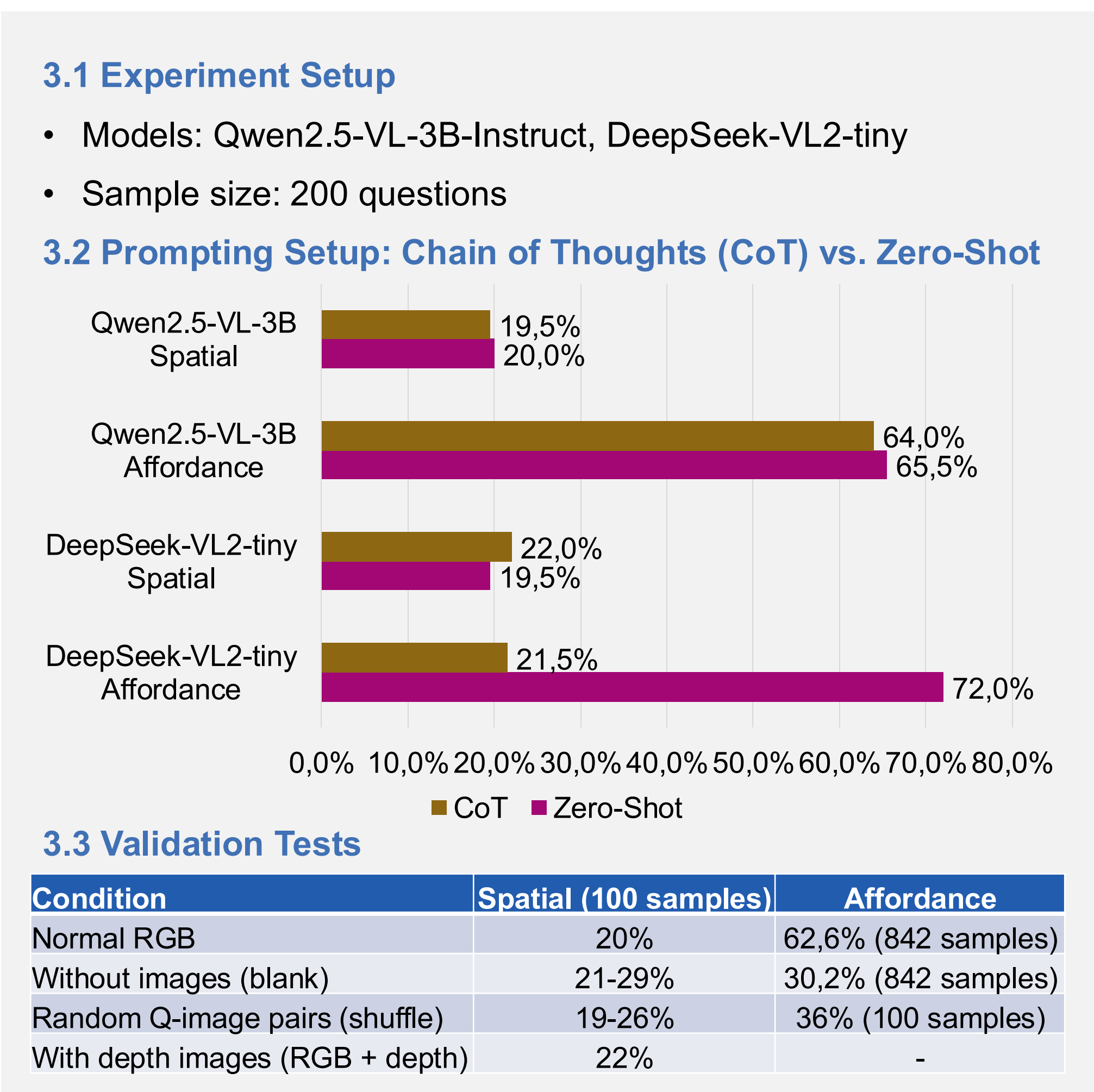
1 Introduction

- VLMs are increasingly integrated into robotics
- Problem:** critical limitations in tasks for physical manipulation
- Our task:** investigate **spatial reasoning & affordance understanding**.

2 Data Categorization



3 Experiments



References

1. K. Chen, S. Xie, Z. Ma, P. R. Sanketi, and K. Goldberg, "Robo2VLM: Visual Question Answering from Large-Scale In-the-Wild Robot Manipulation Datasets," *NeurIPS* 2025, doi: 10.48550/arxiv.2505.15517.

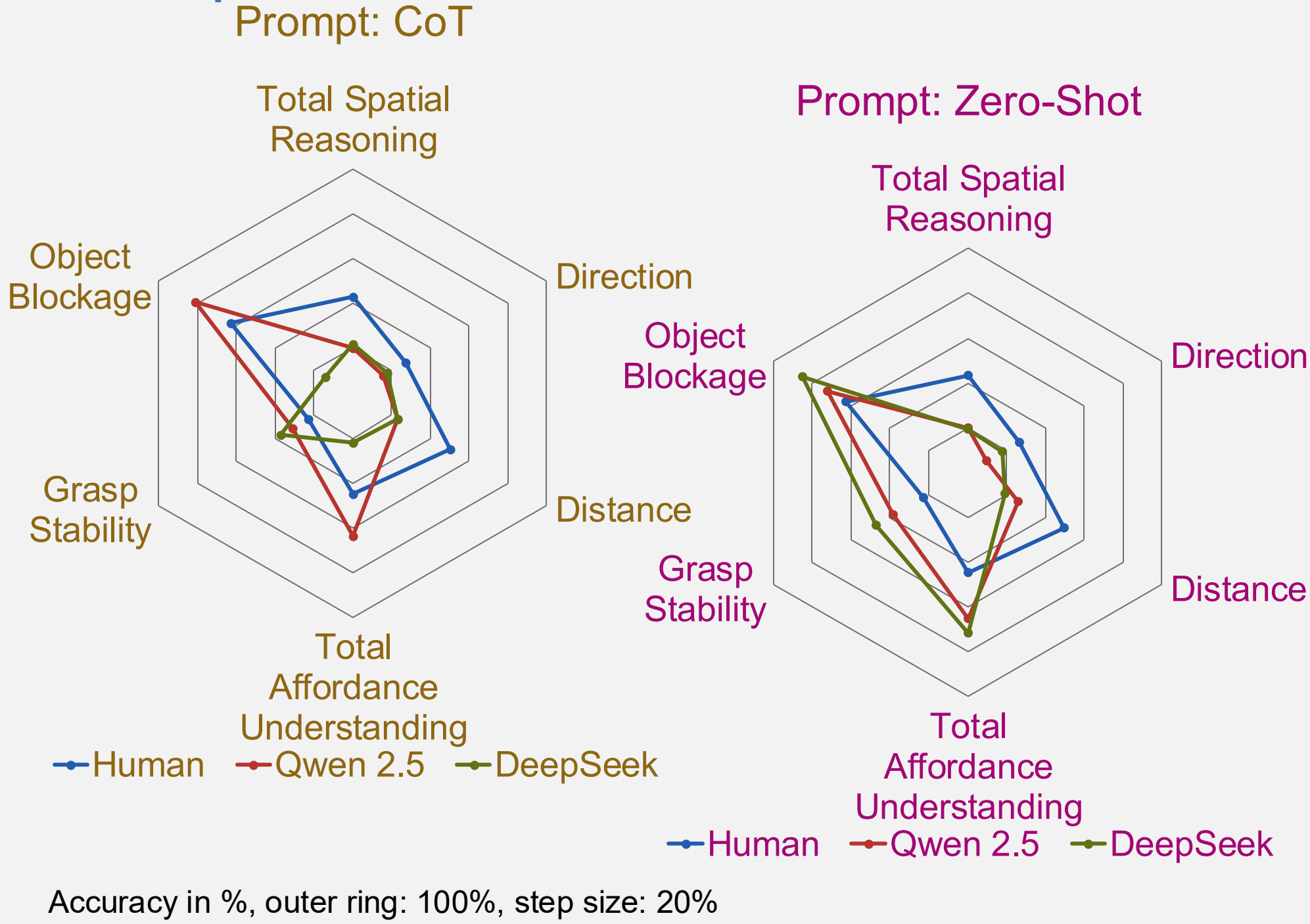
RGB + depth Image Example:



3.4 Comparison to Additional Models

Model	Spatial ZS	Affordance ZS	Spatial CoT	Affordance CoT
SmoVLM-2.2B	20%	23%	66%	47%
Phi-4-multimodal	12%	19%	40%	44%

3.5 Comparison of VLM Performance to Human



4 Discussion

- Limited evidence** that the tested VLMs reliably use depth, camera geometry, and end-effector cues
- CoT** changes model behavior but **not consistently beneficial**
- DeepSeek-VL2 Reasoning Collapse Example:**


Q: Is there any obstacle blocking the robot from reaching bottle?
Expected: A ("No.")
Zero-shot predicted: A, correct
CoT response excerpt: Reason: The presence of an object (bottle) near the end of the robotic arm suggests that it may pose a challenge for the robot's movement.
CoT predicted answer: D ("Yes."), incorrect
- Affordance accuracy is **heterogeneous**
- High task accuracy **DOES NOT** imply visually grounded explanations:

Grounding Stress Test

Image evidence or question prior? contradicts unclear expected

Model	Support correct	Reject wrong	Reject blank	Reject shuffle
Qwen2.5 VL-3B	expected	contradicts	expected	unclear
DeepSeek VL2-tiny	unclear	unclear	expected	expected